

## Data Cleaning Complete Guide (Bangla)

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**Data Cleaning কী?** Raw ডাটা থেকে ভুল, ফাঁকা, duplicate, inconsistent জিনিসগুলো ঠিক করা।

**কারণ:** - Dirty Data = Wrong Model - Clean Data = Better Accuracy

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### Main Steps of Data Cleaning:

#### 1 . Missing Value Handle করা

```
import pandas as pd
df.isnull().sum()
# Drop missing values
df.dropna(inplace=True)
# Fill numerical
df["age"].fillna(df["age"].mean(), inplace=True)
# Fill categorical
df["city"].fillna(df["city"].mode()[0], inplace=True)
```

#### 2 . Duplicate Data Remove করা

```
df.duplicated().sum()
df.drop_duplicates(inplace=True)
```

#### 3 . Wrong Data Type Fix করা

```
df.dtypes
df["date"] = pd.to_datetime(df["date"])
df["price"] = df["price"].astype(float)
```

#### 4 . Outlier Detect & Remove করা

```
import matplotlib.pyplot as plt
plt.boxplot(df["salary"])
plt.show()

Q1 = df["salary"].quantile(0.25)
Q3 = df["salary"].quantile(0.75)
```

```
IQR = Q3 - Q1
```

```
df = df[(df["salary"] >= Q1 - 1.5*IQR) & (df["salary"] <= Q3 + 1.5*IQR)]
```

## 5 . String Data Clean করা

```
df["name"] = df["name"].str.lower().str.strip()
```

## Basic Cleaning Example

```
import pandas as pd
df = pd.read_csv("data.csv")

# Missing value
df["age"].fillna(df["age"].mean(), inplace=True)

# Duplicate remove
df.drop_duplicates(inplace=True)

# Data type fix
df["date"] = pd.to_datetime(df["date"])

# String clean
df["city"] = df["city"].str.lower().str.strip()
print(df.head())

# Check dataset
df.info()
df.describe()
```

**Strategy for Missing Value** Column > 5 0% missing → Drop - Row > 5 % missing → Drop  
Otherwise → Fill

**Steps of Cleaning Data** Load CSV 2. Inspect data & null values 3. Remove missing data  
Fill null values 5. Convert data types 6. Check duplicates 7. Remove duplicates 8. Save cleaned dataset

## Worked Example

```

# Drop unnecessary columns
df.drop(columns=['Notes', 'Salesperson'], inplace=True, errors='ignore')

# Drop rows missing essential info
df.dropna(subset=['Order ID', 'Customer Name', 'Product'], inplace=True)

# Keep only alphabetic City names
df = df[df['City'].str.isalpha()]
df['City'] = df['City'].str.capitalize()

# Convert numeric columns and fill NaNs with median
numeric_cols = ['Quantity', 'Unit Price', 'Total Amount']
for col in numeric_cols:
    df[col] = pd.to_numeric(df[col], errors='coerce')
    df[col] = df[col].fillna(df[col].median())

# Fill text columns with 'Unknown' and strip spaces
text_cols = ['Payment Method', 'Status', 'Category', 'City']
for col in text_cols:
    df[col] = df[col].fillna('Unknown').astype(str).str.strip()

# Convert Order Date safely
df['Order Date'] = pd.to_datetime(df['Order Date'], errors='coerce')
if df['Order Date'].isna().any():
    df['Order Date'] = df['Order Date'].fillna(df['Order Date'].mode()[0])

# Remove duplicates based on Order ID
df.drop_duplicates(subset='Order ID', inplace=True)

# Save cleaned dataset
df.to_csv('messy_sales_data_cleaned.csv', index=False)
print("Cleaned dataset saved as 'messy_sales_data_cleaned.csv'")

```